SDC HACKATHON 2026

Product Requirement Documentation (PRD), updated for the Final

Hosted by Xiamen University Malaysia (XMUM) AI Club. Revised version of the Phase 1 proposal. Final deadline: 1 November 2026

Instructions: This is the revised version of our Phase 1 proposal (Handbook 6.6.4.1). Text that changed since 25 October is marked Updated. The section What changed since Phase 1, and why lists every change with its reason.

1. Team & Project Overview

Team Name	**HRISSA**
Selected Domain	**Domain 2**
Selected SDG Track (1–17)	**SDG 8 – Decent work and Economic growth**
Project / Solution Name	**Gesir.AI (Giser.AI in Phase 1)**
Name (Leader)	**Azer Hamdaoui**
Email & Phone (Leader)	**azer.hamdaoui.07@gmail.com +216 56138490(what'sapp)**
University Affiliation(s)	**Xiamen University Malaysia**
Number of Team Members	**5**

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

Many small and medium-sized enterprises (SMEs) and service-based startups in Malaysia face a common yet often overlooked challenge: the time-consuming back-and-forth communication required to understand customer requests. Managers frequently need to review WhatsApp messages, voice notes, and images, clarify measurements and job requirements, respond to inquiries, and discuss costs before they can begin planning a job. For example, a small painting business may need to exchange multiple messages with a customer just to gather sufficient information to prepare a quotation. With limited staff and resources, these repetitive administrative tasks can quickly overwhelm managers and divert their attention from essential business operations. Economically, the time spent on repetitive communication reduces productivity, delays quotations, increases the risk of losing potential customers, and limits business growth, ultimately affecting profitability and competitiveness. Socially, constant interruptions and excessive workloads contribute to managerial stress and burnout, while delayed responses and uncertainty can weaken customer trust and satisfaction. Environmentally, incomplete or inaccurate job information may lead to incorrect material estimates, unnecessary purchases, excess paint and material waste, and avoidable trips to suppliers or job sites. These impacts are interconnected, as inefficient communication not only consumes valuable time and human effort but also affects business performance, employee well-being, customer experience, and resource utilisation. Therefore, addressing this challenge is essential to helping Malaysian SMEs streamline their operations, improve productivity, support a healthier working environment, deliver more reliable customer service, and adopt more sustainable business practices.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

Gesir.AI is an AI-powered application designed to help Malaysian SMEs, particularly service-based businesses, streamline customer communication and job preparation. By reducing repetitive inquiries and manual information gathering, the application enables business owners and managers to focus on service delivery, improve productivity, and respond to customers more efficiently.

Gesir.AI applies Generative AI (GenAI) to understand customer messages, voice notes, and images, transforming unstructured information into structured, actionable job briefs. Updated: the customer chats with Gesir.AI on the business's WhatsApp number through the WhatsApp Business Cloud API, in English or Bahasa Malaysia. The assistant asks one question at a time, extracts the measurements, surfaces and customer preferences, checks each value against the customer's own words, and asks again when information is missing.

Updated: a deterministic rate engine, never the AI, prices the job from Malaysian market rates, and a scheduler proposes time slots that respect working hours, public holidays and travel time. The owner reviews the estimate in the dashboard or the mobile app and approves or rejects it. Nothing reaches the customer before approval, the server enforces the approver's role, a rejection needs a reason, and every step is written to an activity log.

Updated: we used the skills from the SDC workshops (GenAI API integration, prompt engineering, database architecture, cloud hosting and workflow automation) to build and deploy the product. It is hosted on Replit with a PostgreSQL database, and its three models run on RunPod serverless GPUs.

Ultimately, Gesir.AI aims to help Malaysian SMEs operate more efficiently by reducing communication delays, improving customer satisfaction, minimising estimation errors, and supporting more responsible use of materials and resources.

2.3 What changed since Phase 1, and why

- **Voice calls.** Vapi was replaced by LiveKit and the calls feature is experimental, so it is not part of the demo. Text, photos and voice notes on WhatsApp are the core.
- **Calendar.** Google Calendar sync was replaced by our own scheduler. It books slots with travel time, public holidays and no double booking, and keeps the data in our own database.
- **AI stack.** PyTorch and Hugging Face serving became vLLM workers on RunPod: Qwen2.5-7B-Instruct for chat, Qwen2.5-VL-7B-Instruct for photos, and Faster Whisper for voice notes.
- **Human control.** Approval is now enforced in the backend with three roles (Admin, Approver, Reviewer). A rejection needs a reason and sends nothing to the customer. This follows the Domain 2 requirement that people stay in control.
- **Privacy.** Customers agree to a PDPA notice before anything is read, can erase their data by sending STOP, and the owner can erase a customer.
- **Second trade and reuse.** Aircon servicing was added next to painting with its own rules file and rate card, to show that a new business is a configuration change, not a rebuild.
- **Orbyte fit.** The dashboard now follows Orbyte's layout (Pending Actions, Clients, Projects, Users, Wallet) and the server emits signed events, so it can sit inside the Orbyte hub.
- **Owner access.** A mobile app with push notifications and a real sign-up and sign-in (the owner picks painting or aircon) replaced the single shared password.

2.4 Target users, scope and key features

- **Target users.** Owners and small teams of Malaysian service businesses, such as painting and aircon contractors, who take every enquiry on WhatsApp and have no time for formal training.
- **In scope.** One use case: turning a WhatsApp enquiry into an approved quotation and a booked visit. Interior walls and ceilings are priced for painting; aircon covers servicing by unit count, service type and horsepower.
- **Out of scope.** Real payments, statutory filings, and legal or financial advice. Trim, doors and exterior work go to a person. Only synthetic data is used.
- **Key features.** Chat in English and Bahasa Malaysia; photo and voice-note understanding; deterministic estimates; role-based approval and rejection; activity log; area-aware scheduling; PDPA consent and erasure; dashboard, mobile app and sign-in; signed events for Orbyte; a second trade through configuration.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

(appendix attached at the end)

1. Communication & Frontend Interfaces

- **WhatsApp Business Cloud API:** The main channel where clients chat with us, share their room photos, and lay out their painting needs.
- **LiveKit (experimental):** Answers WhatsApp calls with a voice agent that uses the same assistant. It replaced Vapi in our plan and is not part of the demo.
- **Contractor Web Dashboard and mobile app:** A Replit-hosted dashboard in Orbyte's layout and an Expo mobile app. Owners sign up and sign in, review estimates, approve or reject them, and confirm schedule slots.

2. Backend & Core Frameworks

- **FastAPI:** The engine connecting everything behind the scenes—handling webhooks, routing AI requests, and running our core calculation logic.
- **Estimate PDF:** A one-page estimate letter written by our own code (no PDF library) shows the job and the total, never the price breakdown.
- **Uvicorn:** Runs our FastAPI services reliably with minimal overhead.

3. AI & Compute Infrastructure

- **RunPod:** Runs three serverless GPU endpoints: Qwen2.5-7B-Instruct for chat, Qwen2.5-VL-7B-Instruct for photos and Faster Whisper for voice notes. Prices never come from a model.
- **vLLM and faster-whisper:** The serving stack on RunPod that loads and runs the models.

4. External Services & Productivity APIs

- **Own scheduler:** Replaces Google Calendar. It suggests area-aware slots with travel gaps, never double books, and skips weekends and public holidays.

5. Database & State Management

- **SQL Database (SQLite / PostgreSQL):** Keeps a reliable log of chat histories, contractor rate cards, draft estimates, and customer booking details.
- **Replit Cloud:** The live hosting environment running our backend APIs, contractor dashboard, and database setup in one place.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

 RunPod (cloud GPU, US$25 team credit): Hosts the chat, photo and speech models while keeping language interpretation separate from pricing and scheduling logic.

 Replit (Replit Core for each participant): Hosts the FastAPI backend, the PostgreSQL database and the dashboard at gesir-ai.replit.app.

 KrackedDevs (AI tools through OpenCode): Used while writing and reviewing code. It is not part of the running product.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

Our team applied **design-thinking methodology** to develop Gesir.ai through three key steps. **First, we identified** the challenges faced by painting SMEs, including scattered WhatsApp enquiries, time-consuming quotations, and missed follow-ups. **Second, we designed and prototyped** an AI assistant that extracts job details, identifies missing information, and prepares quotation drafts using predefined pricing rules and human approval. **Finally, we plan to test and improve** the MVP using realistic scenarios and feedback from potential users to evaluate accuracy, usability, and processing time. By Phase 2, our goal is to deliver a technically viable, user-friendly solution with appropriate privacy safeguards, verified integration requirements, and a clear deployment plan for real-world use. *Updated for the Final: we tested the assistant with about 200 stress conversations against the live model and 607 automated tests, and fixed every wrong reply we found. We have not yet tested with real SME owners; that is the first step of our pilot plan.*

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess [Innovation and Creativity as well as Feasibility of build]. Vague or unsupported projections may weaken your case for advancing to Phase 2.

Metric Category	**Your Response**
Target SDG Sub-Target(s)	SDG 8: Decent work and economic growth SDG 8.1: Economic Growth SDG8.2 Diversification and Innovation SDG 8.3: Enterprise Support

Metric Category	**Your Response**
Identify the exact UN SDG sub-target your project addresses *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	- Gesir.AI Targets 8.2 (Diversification and Innovation) which Boost productivity via technological upgrades, innovation, and diversification into high-value and labor-intensive sectors.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Statistics show: - Small business owners typically spend **10 to 15 hours per week** answering support queries themselves, which translates to roughly $19,500 per year in lost productivity (Time burden) (Ringly.io,2026) - Small business owners typically spend **10 to 15 hours per week** answering support queries themselves, which translates to roughly $19,500 per year in lost productivity (Budget Allocation) (Ringly.io,2026)
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	The amount an SME spends varies significantly by how they handle communication: • **Phone support:** $6 to $8 per interaction • **Email and chat:** $3 to $5 per interaction • **AI voice/chat agents:** Roughly $0.40 per call (Ringly.io,2026) ⇨ providing a 90%–95% cost reduction
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	- **Time Savings Focus:** "Cutting the time small business owners spend on manual support queries by **85%**, translating to roughly 10 to 12 hours saved per week per business." Updated result (modelled, not yet measured with a live SME): owner time per enquiry falls from 37.5 to 3.25 minutes, a 91.3% reduction and about 11.4 hours saved a week against the 85% target, at an assumed 20 enquiries a week. Assumptions and the reproducible calculation (scripts/benefit_model.py) are in the Final report, section 4.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

☐ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

☐ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

☐ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

☐ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☐ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☐ Yes ☐ No

Name	Type	License
Qwen/Qwen2.5-7B-Instruct	Chat model (RunPod, vLLM)	Apache-2.0
Qwen/Qwen2.5-VL-7B-Instruct	Photo model (RunPod, vLLM)	Apache-2.0
faster-whisper (SYSTRAN)	Speech recognition code	MIT
Whisper small and base	Speech recognition model	MIT
FastAPI, Uvicorn, Pydantic	API framework and server	MIT, BSD-3-Clause

vLLM (RunPod worker)	Model serving	Apache-2.0
LiveKit Agents and Cloud; Deepgram, Cartesia (via LiveKit)	Voice calls (experimental)	Apache-2.0; provider terms
React, Vite, TypeScript, Expo / React Native	Dashboard and mobile app	MIT, Apache-2.0
psycopg 3	PostgreSQL driver	LGPL-3.0

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☐ No

If yes, please describe the nature of the agreement and which member it concerns:

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7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



1 November

2026